



Searching

1. concept of searching
2. Linear Search.
3. Binary Search.

Searching is the process of finding a given value position in a list of values. It decides whether a search key is present in the data or not. It is the algorithmic process of finding a particular item in a collection of items. It can be done on internal data structure or on external data structure.

There are two types of searching methods available

1. Linear Search
2. Binary Search

Sequential (Linear)Search

- Sequential search is also called as Linear Search.
- Sequential search starts at the beginning of the list and checks every element of the list.
- It is a basic and simple search algorithm.
- Sequential search compares the element with all the other elements given in the list. If the element is matched, it returns the value index, else it returns -1.

Binary Search

- Binary Search is used for searching an element in a sorted array.
- It is a fast search algorithm with run-time complexity of $O(\log n)$.
- Binary search works on the principle of divide and conquer.
- This searching technique looks for a particular element by comparing the middle most element of the collection.
- It is useful when there are large number of elements in an array.

Linear Search

key=34

```
public int linearSearch(int a[],int size,int  
key)  
{  
    int index=-1;  
    for(i=0;i<size;i++)  
    {  
        if(a[i]==key)  
        {  
            index=i;  
            break;  
        }  
    }  
    return index;  
}
```

i=0	i=1	i=2	i=3	i=4	i=5	i=6	i=7	i=8	i=9
23	13	76	54	-34	34	23	123	49	28
0	1	2	3	4	5	6	7	8	9

Search =

123

5	10	15	20	25	30	35
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Mid=3

5	10	15	20	25	30	35
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25 > 20 Check in upper half of array Find mid
of Upper array

5	10	15	20	25	30	35
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Continue checking